COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	8-27-26	**DATE SUBMITTED:**	8-29-26
NAME:	Masongsong, John Marvin B.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 5.1 **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (*CPE232_yourname*). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issue this command. Did something happen? Why?

```
masongsong@workstation:~/CPE212_Masongsong$ git pull
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 710 bytes | 118.00 KiB/s, done.
From github.com:JohnMarvinMasongsong/CPE212_Masongsong
   acd9f75..7fc6859  main       -> origin/main
Updating acd9f75..7fc6859
Fast-forward
 ansible.cfg         |  4 ++++
 install_apache.yaml | 17 +++++++++++++++++
 inventory.yaml      | 16 ++++++++++++++++
 3 files changed, 37 insertions(+)
 create mode 100644 ansible.cfg
 create mode 100644 install_apache.yaml
 create mode 100644 inventory.yaml
masongsong@workstation:~/CPE212_Masongsong$
```

- The changes that I made in github last HOA was pulled into the VM. This is because there are 2 ssh keys that are connected in my github, one from the lab and one from my home, and the changes I made at home was connected to the github and I was able to get said changes in our lab.

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): *ansible-playbook --ask-become-pass install_apache.yml*. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

```
TASK [update repository index] ***********************************************
*
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [centos1]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg":
 "[Errno 2] No such file or directory: b'apt-get'", "rc": 2, "stderr": "", "std
err_lines": [], "stdout": "", "stdout_lines": []}
changed: [ubuntu1]
changed: [ubuntu2]

TASK [install apache2 package] ***********************************************
*
changed: [ubuntu1]
changed: [ubuntu2]

TASK [add PHP support for apache] ********************************************
*
```

3. Edit the *install_apache.yml* file and insert the lines shown below.

```
  GNU nano 7.2                    install_apache.yaml
---
- hosts: all
  become: true
  tasks:


  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"
```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
TASK [install apache2 package] ************************************************
*
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

TASK [add PHP support for apache] *********************************************
*
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

PLAY RECAP ********************************************************************
*
centos1                    : ok=1    changed=0    unreachable=0    failed=0
skipped=3    rescued=0    ignored=0
ubuntu1                    : ok=4    changed=1    unreachable=0    failed=0
skipped=0    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=1    unreachable=0    failed=0
skipped=0    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

- Everything is now installed and centos is skipped because there is no "dnf" command in the playbook.

If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

- name: update repository index
  apt:
    update_cache: yes
  when: ansible_distribution in ["Debian", "Ubuntu]

Note: This will work also if you try. Notice the changes are highlighted.

4. Edit the *install_apache.yml* file and insert the lines shown below.

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
      stae: latest
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache2 package
    dnf:
      name: httpd
      state: latest
    when: ansible_distribution == "CentOS"

  - name: add PHP support for apache
    dnf:
      name: php
      state: latest
    when: ansible_distribution == "CentOS"
```

Make sure to save and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
TASK [update repository index] ***********************************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [install apache2 package] ***********************************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
changed: [centos1]

TASK [add PHP support for apache] ********************************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
changed: [centos1]

PLAY RECAP *******************************************************************
*
centos1                    : ok=4    changed=2    unreachable=0    failed=0
skipped=3    rescued=0    ignored=0
ubuntu1                    : ok=4    changed=1    unreachable=0    failed=0
skipped=3    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=1    unreachable=0    failed=0
skipped=3    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

- The part that I added instead ignored the ubuntu vm since it used a “dnf” package.

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It’s because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

   5.1 To activate, go to the CentOS VM terminal and enter the following:
   systemctl status httpd
   The result of this command tells you that the service is inactive.

```
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: inactive (dead)
       Docs: man:httpd.service(8)
~
~
```

5.2 Issue the following command to start the service:

sudo systemctl start httpd

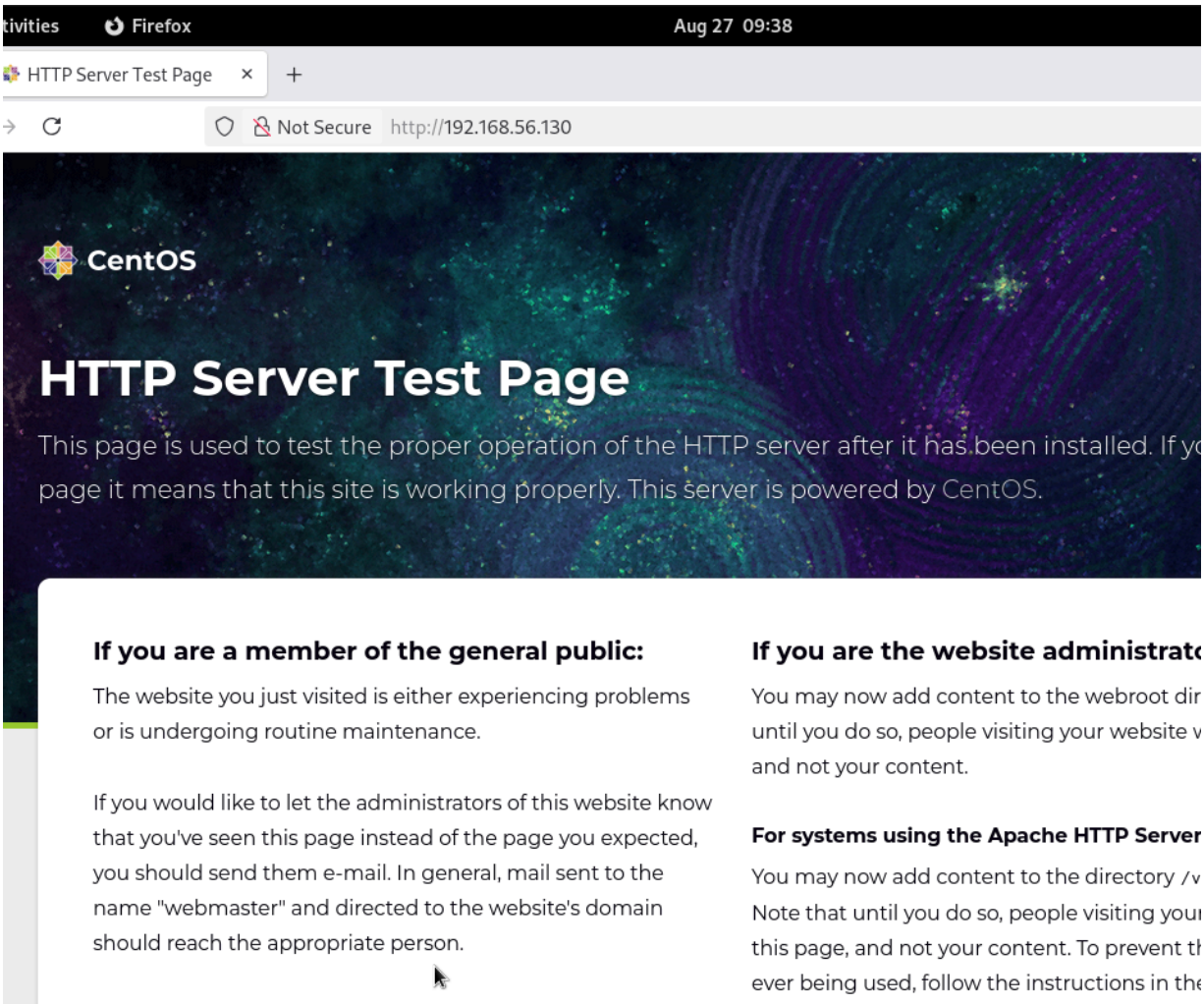
(When prompted, enter the sudo password)

sudo firewall-cmd --add-port=80/tcp

(The result should be a success)

```
[masongsong@centos ~]$ sudo systemctl start httpd
[sudo] password for masongsong:
[masongsong@centos ~]$ sudo firewall-cmd --add-port=80/tcp
success
[masongsong@centos ~]$
```

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)



Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also makes run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook *install_apache.yml*. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index Ubuntu
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index for CentOS
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

   Make sure to save the file and exit.

   Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
skipping: [centos1]
changed: [ubuntu1]
changed: [ubuntu2]

TASK [install apache2 and php packages for Ubuntu] *****************************
*
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [update repository index] *************************************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [install apache2 and php packages for CentOS] *****************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY RECAP *********************************************************************
*
centos1                    : ok=3    changed=0    unreachable=0    failed=0
skipped=2    rescued=0    ignored=0
ubuntu1                    : ok=3    changed=1    unreachable=0    failed=0
skipped=2    rescued=0    ignored=0
ubuntu2                    : ok=3    changed=1    unreachable=0    failed=0
skipped=2    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

- The output is still the same but with a shorter format, meaning we are able to get the same result with shorter commands/shortcuts.

2. Edit the playbook *install_apache.yml* again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidated everything in just 2 plays. This can be done by removing the update repository play and putting the command *update_cache: yes* below the command *state: latest*. See below for reference:

```
---
- hosts: all
  become: true
  tasks:

   - name: install apache2 and php packages for Ubuntu
     apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
     when: ansible_distribution == "Ubuntu"

   - name: install apache and php packages for CentOS
     dnf:
       name:
         - httpd
         - php
       state: latest
       update_cache: yes
     when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
PLAY [all] *********************************************************************
*

TASK [Gathering Facts] *********************************************************
*
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [install apache2 and php packages for Ubuntu] *****************************
*
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

TASK [install apache2 and php packages for CentOS] *****************************
*
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

PLAY RECAP *********************************************************************
*
centos1                    : ok=2    changed=0    unreachable=0    failed=0
skipped=1    rescued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0
skipped=1    rescued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0
skipped=1    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

- The output does not have the update repository index anymore and just does it inside the 2 tasks quietly.

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook *install_apache.yml* again and make sure to follow the below image. Make sure to save the file and exit.

```
---
- hosts: all
  become: true
  tasks:

  - name: install apache and php
    apt:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
PLAY [all] ********************************************************************

TASK [Gathering Facts] ********************************************************
ok: [server1]
ok: [server2]
ok: [centos]

TASK [install apache and php] *************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [centos]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno
: b'apt-get'", "rc": 2, "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines
ok: [server1]
ok: [server2]

PLAY RECAP ********************************************************************
centos                     : ok=1    changed=0    unreachable=0    failed=1    skippe

server1                    : ok=2    changed=0    unreachable=0    failed=0    skippe

server2                    : ok=2    changed=0    unreachable=0    failed=0    skippe


masongsong@workstation:~/CPE212_Masongsong$ nano install_apache.yaml
masongsong@workstation:~/CPE212_Masongsong$
```

- It is not working for CentOS

4. Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place. Edit the *inventory* file and follow the below configuration:

```
192.168.56.120 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.121 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.122 apache_package=httpd php_package=php
```

Make sure to save the *inventory* file and

Finally, we still have one more thing to change in our *install_apache.yml* file. In task 2.3, you may notice that the package is assign as apt, which will not run in CentOS. Replace the *apt* with *package*. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use *apt*, and for CentOS it will automatically use *dnf*. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation: ansible.builtin.package – Generic OS package manager — Ansible Documentation

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
TASK [Gathering Facts] *********************************************************
ok: [server2]
ok: [server1]
ok: [centos]

TASK [install apache and php] **************************************************
ok: [server1]
ok: [server2]
ok: [centos]

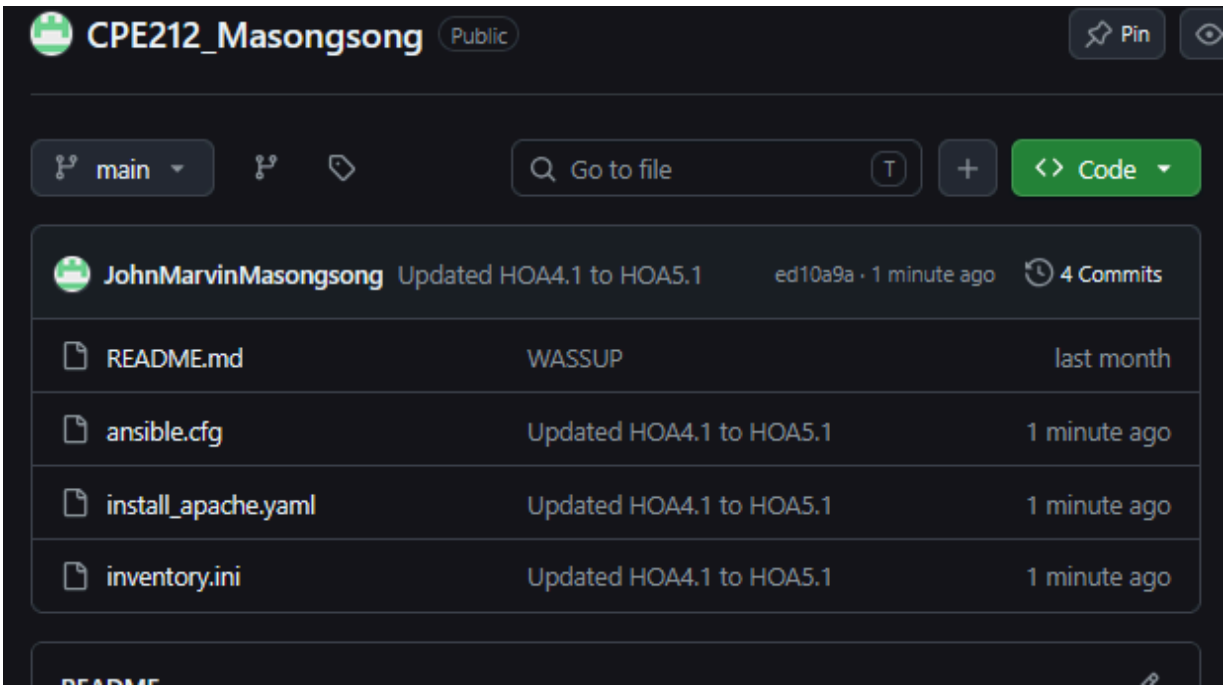
PLAY RECAP *********************************************************************
centos                     : ok=2    changed=0    unreachable=0    failed=0    skippe

server1                    : ok=2    changed=0    unreachable=0    failed=0    skippe

server2                    : ok=2    changed=0    unreachable=0    failed=0    skippe


masongsong@workstation:~/CPE212_Masongsong$
```

- Everything is now working.



- The changes have been pushed and committed to github.

Supplementary Activity:

1. Create a playbook that could do the previous tasks in Red Hat OS.

   Output:

```
PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos]

TASK [install apache and php on CentOS] ****************************************

ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos]

PLAY RECAP *********************************************************************
centos                     : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

[masong@localhost CPE212_Masongsong]$
[masong@localhost CPE212_Masongsong]$
```

Ansible.cfg

```
  GNU nano 5.6.1                          ansible.cfg
[defaults]
inventory = inventory_centos.ini

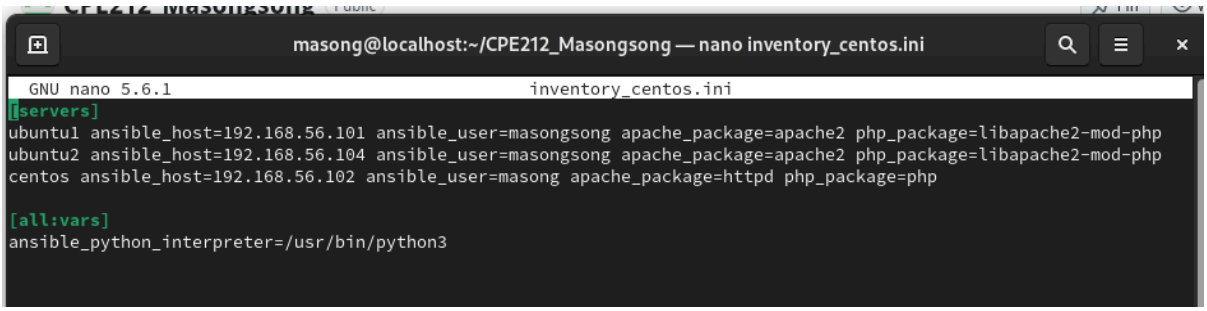
interpreter_python = auto_silent
```

Install_apache_centos.yaml

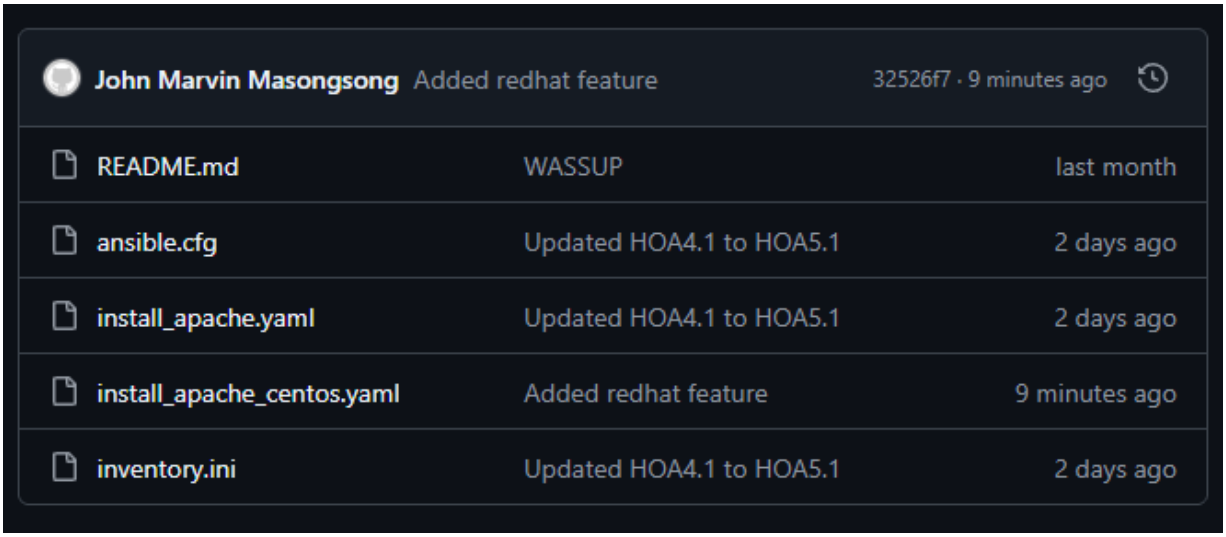
```yaml
---
- hosts: redhat
  become: true
  tasks:


  - name: install apache and php on CentOS
    package:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes
```

Inventory.ini



Github commit



Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **NA**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **NA**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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